

Mathematical Induction – Challenging Questions

1 Define the expression $S(n) = \sum_{p=2}^n \frac{1}{p^2-1}$

(a) Use mathematical Induction to prove that:

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$$S(n) = \frac{(n-1)(3n+2)}{4n(n+1)}$$

(b) Hence, or otherwise, find the value of

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$$\sum_{p=2}^{\infty} \frac{1}{p^2-1}$$

2 Answer the following parts (a) and (b) below:

(a) Prove algebraically that $n(n+1)$ is even for all integers $n \geq 1$

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(b) Hence, use Mathematical Induction to prove that $n^3 + 3n^2 + 2n$ is divisible by 6 for all integers $n \geq 1$.

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3 Answer the following parts (a) and (b) below:

(a) Show that $(k+1)(k+2)(k+3)\dots(k+n)$ is divisible by $n!$, where n and k are positive integers.

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(b) Let m be an arbitrary positive integer.

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Use mathematical induction on n to prove that $(mn)!$ is divisible by $(m!)^n$ for all positive integers m and n .

4 Use Mathematical Induction to prove that for all integers $n \geq 3$.

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$$n^{n+1} > (n+1)^n,$$

- 5 Use Mathematical Induction to prove that for all positive integer values of n , 3

$$\sum_{k=1}^{2^n} \frac{1}{k} < n + 1$$

- 6 Suppose that a and b are positive real numbers 3

Use Mathematical Induction to prove for positive integers n ,

$$\left(\frac{a+b}{2}\right)^n \leq \frac{a^n + b^n}{2}$$

- 7 Prove by Mathematical Induction that for integers $n \geq 4$, 3

$$\ln[(n+2)!] > n + 2$$

- 8 Let n be a positive integer.

- (a) Show that $(1 + \frac{1}{n})^n \geq 2$ 1

- (b) Hence, or otherwise, prove by Mathematical Induction, 3

$$(2n-1)! \leq n^{2n-1}$$

- 9 Using Mathematical Induction, prove that the sum of the interior angles of a polygon with n vertices is $180(n-2)^\circ$ for integers $n \geq 3$. 3

- 10 Prove by mathematical induction that for positive integers n , 3

$$\frac{\sin x}{\cos x + \cos 3x} + \frac{\sin x}{\cos x + \cos 5x} + \cdots + \frac{\sin x}{\cos x + \cos(2n+1)x} = \frac{\tan(n+1)x - \tan x}{2}$$

- 11** Prove by mathematical induction that $5^{2n^2} - 1$ is divisible by 4, for integers $n > 0$. **3**

- 12** Consider the Sum of an Arithmetic Series with common difference d and first term a ,

$$a + (a + d) + (a + 2d) + \cdots + [a + (n - 1)d] = \frac{n}{2} [2a + (n - 1)d]$$

- Prove the statement above for all positive integers n , using Mathematical Induction **3**

- 13** Consider the Sum of a Geometric Series with common ratio r and first term a , **3**

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$$

- Prove the statement above for all positive integers n , using Mathematical Induction. Assume that $r > 1$.

- 14** Prove that $2002^{2n+1} + 2003^{2n+1}$ is divisible by 4005 for all positive integers n . **3**

- 15** Prove by mathematical induction that for integers $n \geq 1$, **3**

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2$$

- 15** Prove by mathematical induction that for integers $n \geq 1$, **3**

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4} [n(n + 1)]^2$$

- 17** Prove using mathematical induction that for all positive integers n , **3**

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n-1} + \sqrt{n}} = \sqrt{n} - 1$$

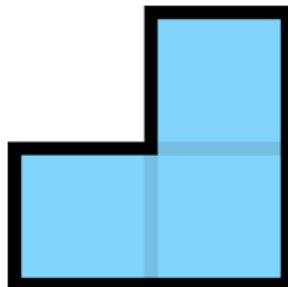
- 18** A circle and a chord of that circle are drawn in a plane. Then a second circle and a chord of that circle, are added. **3**

Repeating this process, once there are n circles with chords drawn, prove that the regions in the plane divided off by the circles and chords can be coloured with three colours in such a way that no two regions sharing some length of border are the same colour.

- 19** If n lines are drawn in a plane, and no two lines are parallel, what is the maximum number of regions that they separate the plane into? **3**

Hint: Determine the number of regions in the first 4 terms and determine a relation between the number of lines n and the number of regions

- 20** Show that for integers $n \geq 0$, a $2^n \times 2^n$ chessboard with one square removed can always be tiled by "L-shaped (Tromino)" pieces as seen below. That is, ONLY pieces formed by removing a corner from a 2×2 square. **3**



Hint: Assume true for $n = k - 1$, and then prove for $n = k$.